

Package Inserts			
I.	Brand or Product Name: TILMICOSIN-LD 250MG/ML Oral Solution		
II.	Name and Strength of Active Substance: Tilmicosin, 250mg in 1mL		
III.	Product Description: A brown clear solution.		
IV.	Target Species: Chicken (broilers and pullet), turkeys, pigs and cattle (calves).		
V.	Pharmacodynamics: Tilmicosin is a mainly bactericidal semi-synthetic antibiotic of the macrolide group. It is believed to affect bacterial protein synthesis. Tilmicosin has a wide spectrum of activity against Gram-positive organisms and is particularly active against Pasteurella, Actinobacillus (Haemophilus) and Mycoplasma organisms of bovine, porcine and avian origin. Tilmicosin has some activity against certain Gram-negative micro-organisms. Cross resistance between tilmicosin and other macrolide antibiotics has been observed. Macrolides inhibit protein synthesis by reversibly binding to the 50S ribosomal subunit. Bacterial growth is inhibited by induction of the separation of peptidyl transfer RNA from the ribosome during the elongation phase. Ribosomal methylase, encoded by the erm gene, can precipitate resistance to macrolides by alteration of the ribosomal binding site. The gene that encodes for an efflux mechanism, mef, also brings about a moderate degree of resistance. Resistance is also brought about by an efflux pump that actively rids the cells of the macrolide. This efflux pump is chromosomally mediated by genes referred to as acrAB genes. Resistance of Pseudomonas species and other Gramnegative bacteria, enterococci and staphylococci may be precipitated by chromosomally controlled alteration of permeability or uptake of the drug.	<u>Pigs:</u> 15-20 mg tilmicosin per kg body weight for 5 days, i.e. 6-8 ml of product for 100 kg body weight corresponding to 80 ml of product per 100 litres of drinking water for 5 days. <u>Chickens:</u> 15-20 mg tilmicosin per kg body weight for 3 days, i.e. 6-8 ml of product for 100 kg body weight corresponding to 30 ml of product per 100 litres of drinking water for 3 days. <u>Turkeys:</u> 10-27 mg tilmicosin per kg body weight for 3 days, i.e. 4-11 ml of product for 100 kg body weight corresponding to 30 ml of product per 100 litres of drinking water for 3 days. <u>Calves:</u> 12.5 mg tilmicosin per kg body weight two times per day for 3-5 days, i.e. 1 ml of product for 20 kg body weight two times per day for 3-5 days. One 1000 ml bottle is sufficient to medicate 1250 liters of drinking water for pigs or 3400 liters of drinking water for broilers, turkeys and pullets. One 1000 ml bottle is sufficient to medicate drinking water or milk replacer for 50–83 calves (40 kg b.w.). One 1000 ml bottle is sufficient to medicate drinking water or milk replacer for 33 calves (60 kg b.w.). Medicated drinking water should be prepared fresh every 24 hours using only clean water. Medicated milk replacer should be prepared fresh every 4 hours using only clean water. If signs of disease do not significantly improve within 3-5 days, the diagnosis should be re-evaluated and treatment changed. To avoid underdosing body weight should be determined as accurately as possible. The uptake of medicated water depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of the product has to be adjusted accordingly Do not administer to pigs in a wet feeding system.	XII. Pregnancy and Lactation: The safety of the product has not been established during pregnancy and lactation. Use only in accordance with risk/benefit assessment by the responsible veterinarian. XIII. Side Effects: None known. XIV. Symptoms and Treatment of Overdose: Pigs drink less water when a dose of 300 to 400 mg/liter (1.5 to 2 times the recommended dose) is administered. Although this will result in less intake of tilmicosin, it might lead to dehydration of the animals. Replace with untreated water when needed. No symptoms were seen in poultry treated at 375 mg/liter during 5 days. A dose of 75 mg/liter during 10 days resulted in less consistent faeces. No symptoms of overdose were noticed in turkeys treated at 375 mg/liter of drinking water during 3 days. No symptoms were noticed at 75 mg/liter during 6 days. Except for a slight decrease in milk intake, no symptoms of overdose were seen in calves treated at 5 times the recommended dose or during twice the recommended treatment period. XV. Withdrawal Period: Meat and offal: Pigs: 14 days Calves: 42 days. Chickens: 12 days Turkeys: 19 days Eggs: Not authorised for use in birds producing eggs for human consumption.
VI.	Pharmacokinetics: When administered orally to chickens, turkeys and pigs with drinking water and to calves with milk replacer, tilmicosin is absorbed and moves rapidly out of the serum into areas of low pH. This results in very low serum concentrations, but detectable levels of tilmicosin are found in lung tissue as early as 6 hours after starting the treatment. In chicken or turkeys, tilmicosin is also detected in pooled air sac tissue as early as 6 hours after starting the treatment. It is also known that tilmicosin is concentrated in alveolar macrophages of swine. When administered orally to calves tilmicosin is detected in lungs after 6 hours and remains at the therapeutic level up to 60 hours from the last dose.	IX. Contraindication: Do not use in case of hypersensitivity to the active substance or in cases of known resistance to tilmicosin. Do not use in horses.	XVI. Storage Condition: Stored at ≤ 30°C / store below 30°C. Protect from direct sunlight.
VII.	Indication: <u>Pigs:</u> For the treatment and prevention of respiratory infections associated with Mycoplasma hyopneumoniae, Pasteurella multocida and Actinobacillus pleuropneumoniae when the disease has been diagnosed at the herd level. <u>Chickens:</u> For the treatment and prevention of respiratory infections in poultry flocks associated with Mycoplasma gallisepticum and Mycoplasma synoviae when the disease has been diagnosed at the herd level. <u>Turkeys:</u> For the treatment and prevention of respiratory infections in turkey flocks associated with Mycoplasma gallisepticum and Mycoplasma synoviae when the disease has been diagnosed at the herd level. <u>Calves:</u> For the treatment and prevention of respiratory infections associated with Mannheimia haemolytica, P. multocida, Mycoplasma bovis and M. dispar when the disease has been diagnosed at the herd level.	X. Warning and Precautions: <u>Special warnings for each target species</u> Tilmicosin should not be administered by injection to pigs. The product contains disodium edetate. The uptake of medicated water can be altered as a consequence of illness. If the uptake is insufficient alternative treatment may be required. <u>Special precautions for use in animals</u> Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin related substances Use of the product should be based on susceptibility testing and take into account official and local antimicrobial policies. Do not allow horses or other equines access to drinking water containing tilmicosin. <u>Special precautions for the person administering the veterinary medicinal product to animals</u> People with known hypersensitivity to tilmicosin should avoid contact with the product. The veterinary medicinal product may cause irritation or sensitisation by skin contact. Avoid skin and ocular contact. Wear protective gloves and protective clothes when handling the veterinary medicinal product. In case of contact with skin or eyes, rinse abundantly with fresh water. If irritation persists and in case of incidental ingestion, seek immediately medical advice or call a poison center (dangers linked to disturbances in cardiac conduction). Wash hands after use.	XVII. Proposed Shelf Life: As packaged for sale: 3 years After first opening of the immediate packaging: 28 days After dilution or reconstitution: 24 hours XVIII. Dosage Forms and packaging available: In liquid form, packed in 1L container. XIX. Name and Address of manufacturer NAM PHARMA SDN. BHD. 5, Lebuhr Perusahaan Klebang 11, Taman Perindustrian Antarabangsa IGB, 31200 Chemor, Perak, Malaysia. XX. Name and Address Product holder KTY CHEMICALS SDN BHD 65, Jalan Cenderawasih 3, Taman Cenderawasih Indah, 14300 Nibong Tebal, SPS, Penang, Malaysia. XXI. Date of Revision of Package Insert: 20/12/2019
VIII.	Recommended Dose and Administration: For oral use only. The product must be diluted in drinking water or milk replacer before administration.	XI. Interaction with other medicinal product: Cross resistance between tilmicosin and other macrolide antibiotics and lincosamides has been observed. Tilmicosin may lessen the antibacterial	